Final project: problem and approach

Final presentations: Friday 23 October 2026 (23/10)

Each group: **15 minutes** of presentation, followed by **5 minutes** of questions from other groups.

Choose a dataset from the UCI Machine Learning Repository.

Choose one approach

1. **Classification:** train and evaluate a decision-tree classifier.
2. **Variable selection:** apply course methods to select useful predictors and make accurate predictions, e.g., forward selection or LASSO.
3. **Combined:** use variable selection with decision-tree classification.

Formulate the problem clearly

- Define the prediction target, the decision to support, and a suitable evaluation metric.
- Describe and cite the dataset: observations, variables, target distribution, missing values, and required preprocessing.

Final project: report and justification

Submit **one properly compiled PDF report** containing:

- **Methods and choices.** Explain your approach. Motivate the model, preprocessing, and hyperparameters, e.g., tree depth, leaf size, or regularization penalty.
- **Evaluation.** Compare with a baseline. Tune by validation or cross-validation, fitting preprocessing and selection within each training split. Report final test performance.
- **Evidence and recommendation.** Include readable code snapshots, labeled plots, and result tables that justify your final model or variable set, recommendation, and limitations.

LLM use: full disclosure in the same PDF

If an LLM helped prepare **any part of the report**, include the **full prompt and full answer, verbatim**, for every relevant exchange, including follow-ups. Put these in an appendix and identify the parts of the report they supported.